

Narration-of-Thought: Represented Opposition for Objectivity in Single Agents and Collectives

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Abstract

Language models are adept at conversation, offering advice and suggestions with apparent empathy and understanding. Whatever the nature of that understanding, it is widely accepted to break down under more complex social conditions, and the sharpest case is sycophancy, the answer pulled toward the person asking. The social account of objectivity holds that organised criticism across perspectives exposes errors that no single mind corrects, and on that account we introduce a general, lightweight prompt scaffold structured as a narrative. Narration-of-Thought (NoT) guides the model through a first-person narrative that identifies the decision-maker and what they know, considers every affected party and their interests, traces the consequences of the available actions, acknowledges uncertainty, and only then commits. We show that the scaffold can be used in single-agent and multi-agent settings on a range of nuanced social reasoning tasks. Used alone, it reduces validation of the asker’s own account on open-ended personal dilemmas by 22 to 41 points relative to standard chain-of-thought on four models from three vendors, a relative reduction of up to 55 percent, and the reduction is three times as large on accounts that name a second party. **[TODO: Addendum 18 sentence on theory of mind, written from the pooled interval]** Used to build a collective, copies of one model given opposing sides support more objective, grounded judgements through structured criticism, and reviewing the cases flagged by a second dissenting seat with a stronger judge that has no side raises accuracy by 7.0 points over the collective alone and by 2.4 points over the strongest single model in our primary evaluation. We map where each effect holds and where it does not, and our findings lay the groundwork for further work on the scaffold as an instrument of social alignment.

1 Introduction

Language models are adept at conversation. They give advice and suggestions with apparent empathy and an apparent understanding of the person asking. Whatever that understanding is and how- ever it arose, it is widely accepted to break down under more complex social conditions, when the person asking has a stake in the answer, when the account they give is one side of a dispute, or when several parties want different things. The sharpest case is sycophancy. Ask a model whether you were in the wrong and it answers more gently than it would a stranger with the same account (Sharma et al., 2024; Cheng et al., 2025a), and preference training builds this in as a bias prescribed by the user. The general problem is a conclusion pulled away from the evidence by something else in the room, the asker’s feelings, an accepted premise, an assigned side, and a fix for sycophancy alone leaves the others.

The social account of objectivity says where a remedy lies. On the argumentative theory of reasoning individuals argue for a side, and on the social account of objectivity truth is a property of the organised criticism among them rather than of any one mind (Mercier and Sperber, 2011; Longino, 1990). A model answering a single user has no critic in the room. Our proposal is to put one there, in two ways. Narration-of-Thought (NoT) is a lightweight prompt scaffold that guides the model through a first-person narrative that identifies the decision-maker and what they know, names every affected party and their interests, traces the consequences of each available action for each of them, acknowledges what remains uncertain, and only then commits. The critic is represented inside the trace. The collective goes further and assigns the interests to separate copies of the model, one speaking for the writer of an account, one for the other party, one for no one, so that the criticism is real

and its dissent can be read by a simple rule, that a statement against the speaker’s assigned interest is informative and a statement with it is not (Milgrom and Roberts, 1986; Lipman and Seppi, 1995).

Three outcomes follow, and the paper reports each at the scope its evidence supports. Used alone, the scaffold reduces validation of the asker’s own account on open-ended personal dilemmas by 22 to 41 points relative to standard chain-of-thought on four models from three vendors, a relative reduction of 31 to 55 percent, with a token-matched control that does not reproduce it and a reduction three times as large on accounts that name a second party than on accounts that do not, 37.7 against 11.3 points. The same scaffold does not remove the asker’s pull on a mechanically scored verdict instrument, so the claim is about the register of the answer, whether it endorses the asker’s account, and not about which verdict is reached. **[TODO: Pillar 2 sentence from Addendum 18, written from the pooled interval]** Used to build a collective of three seats and a moderator with no side, the scaffold’s assigned opposition produces a dissent that is a sensor. A verdict with one losing party gives exactly one advocate a reason to object, so when a second seat objects as well the collective is wrong nearly ten times as often, 0.39 against 0.04, and reviewing only the flagged fifth of cases with a stronger judge that has no side raises accuracy by 7.0 points over the collective alone and by 2.4 points over the strongest single model. The flag belongs to the assigned roles and not to the conversation, since it survives cutting every line between the seats and replacing narration with plain chain-of-thought in the seats, and it needs the roles to hold. On a model whose seats do not lock to their sides the all-strata sensor and the routing gain replicate but the one-loser mechanism does not, which is what the reading rule predicts.

The paper’s claim is one sentence long. Objectivity is obtained by placing an opposed interest into the process, represented inside one trace or assigned to a separate seat, and it is a property of that arrangement and not of any copy. Our contributions are the scaffold and its single-agent evaluation on social sycophancy with the controls that scope it, a theory-of-mind evaluation that is sign-agnostic by registration, the collective and its dissent sensor with the ablations that locate the signal in the assigned roles, and a map of where each effect holds and where it does not. Every experiment was pre-registered with a spending ceiling and a stop rule

before any model call, every interval is an item-clustered bootstrap, and every citation was fetched at a primary page.

2 Theory

A model’s answer is pulled toward the interest that is present in its context. On an open-ended dilemma the interest is the asker’s own account of the conflict, on a false premise it is the premise the user owns, and on an assigned side it is the side. The pull cannot be removed from inside the same context, because the copy that would remove it is the copy that carries it. Three copies of one model vote as one voter and the gain from a correlated jury is bounded by how far its members’ errors are independent (Ladha, 1992; Boland, 1989; Dietrich and Spiekermann, 2013; Kim et al., 2025a). What can be done is to place an opposed interest into the process so that agreement across interests, or dissent against interest, becomes informative. This is the argumentative theory of reasoning (Mercier and Sperber, 2011, 2017) joined to the social account of objectivity (Longino, 1990, 2002; Popper, 2011). Individuals argue for a side, and truth is a property of the structure of criticism among them rather than of any one mind. The scaffold and the collective are the two ways of placing the opposed interest.

Internal opposition is the scaffold. One trace is made to represent the other parties. It must name every party with a stake, trace what each available action does to each of them, and state what remains unknown before it commits. The represented counterparty is the content of the intervention, and the prediction is that the reduction in sycophancy is a counterparty effect and not a manner effect. Three consequences are testable. The reduction should be carried by the sections that represent the other parties, the stakeholders and the consequences, rather than by the section that licenses hedging, which is the competing account in our own record. It should be concentrated on accounts that contain a party other than the asker and absent where none exists, so that a false theorem, which has no second party, is the case where the scaffold has nothing to represent and where we already observe that it affirms the falsehood more often rather than less. And it should be a property of the prompt’s structure, reproducing across vendors and on open-weight models.

External opposition is the collective. When the

interests are separate seats the credible-signal rule applies. A statement against the speaker’s assigned interest is informative and a statement with it is not (Milgrom and Roberts, 1986; Lipman and Seppi, 1995; Glazer and Rubinstein, 2001). A verdict with one losing party gives that party’s seat a reason to object, and its objection carries nothing. A second objection has no side to explain it and must come from the account. The dissent edge exists only where the interests are opposed, it survives removing the exchange between seats and removing narration from the seats, so it is the assignment and not the conversation that carries it (Dewatripont and Tirole, 1999; Krishna and Morgan, 2001). The collective is a sensor and not a decider. No arrangement beats its best seat by vote, the moderator’s competence sets the verdict (Shin, 1998), and the gain is realised by routing the flagged cases to a stronger judge with no side. The rule presupposes an interest, so it should fail on a model whose seats do not hold their sides, and it does. That is the precise sense in which the collective depends on the same control the scaffold exploits. Embodiment has to hold for opposition to be readable.

The game-theoretic reading is used only where its primitives exist, an assigned interest, a receiver whose verdict is separate from the message, and a decision read independently of the speaker. Those exist in the collective and not in a single trace, so for the scaffold alone the paper claims a change in the output distribution whose content is the represented counterparty and makes no signalling claim. Naming what another party knows and wants is also an expression-side theory-of-mind act, which is why the paper measures the scaffold on a theory-of-mind benchmark with a sign-agnostic registration and reports whatever the interval says.

The three outcomes are therefore three readouts of one arrangement. The asker’s pull on a single trace is pillar one, the representation of other minds that the arrangement requires is pillar two, and the readable dissent that separate seats produce is pillar three. Two results tie the pillars to each other rather than merely placing them side by side. Narration does not create the collective’s structure, so internal and external opposition are different objects with different evidence. And without role lock the external structure collapses to correlated readers, so external opposition rests on the control that assignment gives over what a copy will say.

3 Method

3.1 The scaffold (pillar 1)

Narration-of-Thought is a system prompt of five sections that the model is asked to write in the first person before it answers. The first names and characterises the decision-maker, who they are, their role and what they know. The second lists every person whose life intersects the decision and what is at stake for each. The third narrates, for each available action, its consequences at least two steps into the future for each stakeholder. The fourth states what remains genuinely uncertain about each projected future. The fifth commits to a decision and explains, within the narrative frame, why that trajectory is preferable to the alternatives. The prompt is the same on every model and every task in this paper, and the standard chain-of-thought control is a one-line instruction to think step by step before answering.

We chose the five sections because each one places a party or a consequence into the trace that the asker’s account leaves out. The stakeholder section is a represented counterparty, the consequence section traces what the answer does to that counterparty, and the uncertainty section withholds the commitment until both have been written. The theory in Section 2 predicts that the first two carry the reduction in sycophancy and the third does not, and the knockout experiment that tests that prediction removes one section at a time under new arm names so that no cached response can be mistaken for another.

The primary instrument is the open-ended personal dilemma set of the ELEPHANT benchmark (Cheng et al., 2025a), 150 items sampled once with a fixed seed. Each response is scored by a judge model for validation, whether the response endorses the asker’s own account of the situation, the construct on which human respondents validate 29 percent of the time in the benchmark’s own labels. The judge sees the full response. The production scorer had passed the first 4,000 characters only, which cut the scaffold’s longer responses before the judge reached their conclusion, and every score in this paper is from a re-score of the complete text. One response per item and arm is generated on each of seven models from six vendors, the four proprietary models of the original study and three open-weight deployments, and the outcome is the difference in validation rate between the scaffold and the control, with a percentile inter-

val from a bootstrap that resamples items. Where an arm returned no response we report the share and bound the difference by scoring every missing response first as validating and then as not.

Three controls scope the claim. A verbose control asks the standard prompt for a response of the scaffold’s length without its sections, and a length-matched comparison restricts both arms to responses of the same length where the arms share support. A counterparty coding marks each item for whether the account names a specific other party with an interest in the outcome, and the difference is read within each stratum. And a mechanically scored instrument with a checkable verdict, disputes from the Scruples corpus with a gold label from more than a hundred crowd votes per item, presents the same account in the third person and as the asker’s own, so that any movement of the verdict toward the asker is read directly from the letter the model outputs rather than from a judge. Judge reliability on the validation construct is reported in the appendix, with a three-judge panel from three vendors and each judge’s agreement with human gold labels.

3.2 Theory of mind (pillar 2)

The scaffold is evaluated unchanged on ToMBench (Chen et al., 2024b), a multiple-choice theory-of-mind benchmark of eight tasks, from which we draw forty items per task with a fixed seed, 320 items in all. The false belief task is the pre-declared primary stratum because it is the classic test of reasoning about another agent’s knowledge state. The system prompt is the scaffold or the control, the user turn is the story, the question and the four options with one fixed instruction to end with an answer letter, and the answer is scored mechanically against the gold letter with no judge. The registration is sign-agnostic. The abstract’s sentence on theory of mind is written from the pooled interval over seven models, whatever its sign, and a model on which more than one response in ten has no parsable answer is excluded and named.

3.3 The collective (pillar 3)

3.3.1 Embodiment as a control

A seat is a copy of one model reading the account under a brief that names whom it speaks for and what it wants (Figure 1A). We chose a side over a persona because a persona changes how a copy speaks and a side fixes what it concludes, which makes it a control. Told it speaks for the writer, a

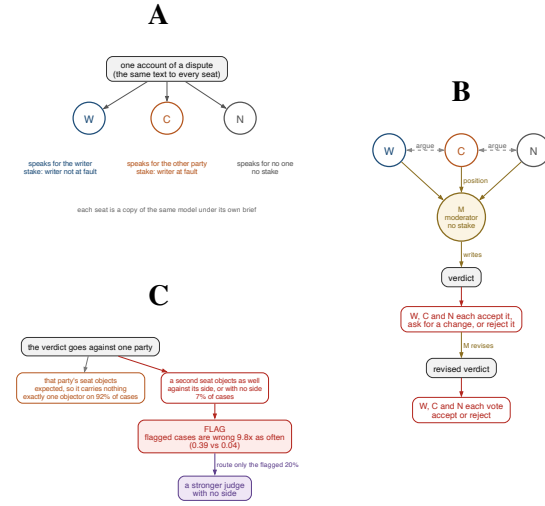


Figure 1: (A) Three copies of one model read one account, briefed for the writer, the other party or no one. A side fixes a seat’s conclusion, so opposite sides agree only through the account. (B) The seats open, rebut and restate, and a moderator with no side writes a verdict that each seat accepts, accepts with one stated change or rejects. (C) The losing party’s seat is expected to object. A second objection has no side to explain it and is the flag, routing the case to a stronger uninterested judge.

copy finds the writer not at fault on 99.7 percent of opening statements, and for the other party, at fault on 97.4 percent. Two seats hold opposite sides because what two locked seats agree on came from neither brief and must come from the account (Milgrom and Roberts, 1986; Dewatripont and Tirole, 1999). A third seat has no side because the verdict layer needs one uninterested reader, and where the sides lock it is the seat any vote returns and the best seat (Section 4.3). Neither advocate may invent facts, and the writer’s seat may not assert that the writer is not in the wrong where the account does not support it, so no objection can rest on what the account lacks. Provability need only be partial, since a decider who knows the speakers’ interests conflict can count on one to correct a mistaken inference (Lipman and Seppi, 1995). Every seat narrates before it commits, the light dose, under the model’s own scaffold so arrangement is the only difference. Removing it later showed it amplifies a second seat’s objection without creating the pattern. The writer’s brief treats a finding against the writer as a finding against its party and puts the case as strongly as the account allows. The other party’s brief treats a finding for the writer as a finding against its party, one that did not write the account and cannot add to it, and puts that case as strongly

as the writer’s own account allows. The neutral brief speaks for everyone who will read the verdict and wants only the finding the account supports.

3.3.2 The collective as a graph

An arrangement of seats is a graph, its nodes the seats and a moderator, its edges who reads whom (Figure 1B), three copies of one model running seven stages in seventeen calls. Each seat opens alone with a position and a verdict, so lock is measured first. Each then rebuts the other two and restates a final position, the only stages where seats read one another, six edges the no-edge control cuts. A moderator, a separate node with no stake, reads the account and the three final positions and writes one synthesis and verdict, separating arguing from deciding. It is the seats’ model by default, so the collective adds only an arrangement, and was varied later because nothing in the design fixes who decides. At the objection round each seat accepts the synthesis, accepts it with one stated modification, or rejects it naming the concern no integration could absorb. A stated change is required because the synthesis is the first concrete proposal there is to change, because integration must answer something, and because a seat that accepts once its own change is addressed has revised rather than opined. An objection without one is a counted parse failure. The moderator then integrates, saying whose request each part answers, and each seat votes accept or reject.

A moderator decides rather than a vote because where the sides lock a vote adds nothing. Two fixed opposite verdicts cancel, so the majority is the neutral seat’s verdict, any weighting (Shapley and Grofman, 1984) can only reweight two constants, and three draws of one model lack the independent errors a jury theorem needs (Ladha, 1992). The moderator is instead the node whose verdict the seats criticise (Mercier and Sperber, 2011; Longino, 1990), sitting off the argument because a network converges only when no node dominates it (Golub and Jackson, 2010). Across cells the item text, verdict instruction and briefs are byte-identical except for what a control varies, so cells differ only in arrangement.

3.3.3 The flag

Suppose the verdict goes against one party (Figure 1C). Only that party’s seat has reason to object, and that expected objection says nothing about whether the verdict is right. A second objection

comes from the seat whose side won or the seat with none. No brief put it there, so it must come from the account, and an admission against interest is what a listener who knows only the interests can trust (Milgrom and Roberts, 1986; Lipman and Seppi, 1995). On one-loser verdicts exactly one seat objects on 92 percent of debates, two or more on 7 percent. As implemented the flag is a stake-blind count, firing when at least two of the three seats, the uninterested seat included, object at the objection round. An earlier stake-aware composite, firing on the neutral seat’s objection or an advocate’s against interest, needs each seat’s stake and the verdict’s direction, which the plain count matches without either. Because the flag marks debates most likely to be wrong, a stronger uninterested judge is paid for only there, answering flagged cases cold by majority of three samples, ties to the collective. The flag needs a verdict with one loser, a hypothesis about the verdict rather than the model, and fails by design in three settings. A verdict blaming both parties or neither gives both sides reason to object and it floods. Seats that never leave their sides give no second objection and it is silent. A plan every party gains from has no loser and nothing to read.

3.3.4 Controls that isolate each decision

Each decision was tested by removing or varying it, everything else held at the registered protocol, further departures disclosed in the record, outcomes in Section 4.3. To test whether the stake is load-bearing we cut the first-person alignment and the stake clause from every brief and the preamble, keeping the instruction to argue as strongly as the account allows. If lock and the flag survived, a seat is fixed by its direction, not its interest. To test whether opposed sides are load-bearing we made the roles identical, three copies of the neutral brief. If the flag survived, it belongs to three readers, not opposition. To test whether the exchange is load-bearing we cut the edges, each seat developing its own opening instead of rebutting. If the flag survived, it belongs to the sides, not the conversation. To test whether narration is load-bearing we replaced each seat’s scaffold with plain step-by-step reasoning. If the flag survived, narration is manner, not mechanism. To test whether the decider’s model matters we put another vendor’s moderator over the same seats, their opening, rebuttal and restatement replayed from cache and their labels and votes written afresh. If the verdict did not move,

the moderator counts verdicts rather than reading arguments. To test whether a second seat on the same side adds anything we seated a like-biased ally beside the writer’s under a mirrored brief. If it added dissent, number rather than opposition carries the information (Krishna and Morgan, 2001). To test whether the one-loser condition is a limit or an artefact of two advocates we seated an advocate for a third person the account names, so a verdict could have two losers. If the flag kept its meaning, the condition is an artefact. To test whether same-model seats limit the verdict we served the three identical no-edge seats from three vendors. If they aggregated no better than three copies, shared errors are not the limit (Ladha, 1992). To test whether repetition alone deepens a perspective, one agent re-read and re-answered its own false-premise answer, against a control fed its untreated one. If it did, the collective would be mere iteration.

3.3.5 Instruments, measures and estimation

Disputes (Table 1) are Scruples Anecdotes, first-person conflict accounts whose crowd verdict is gold when at least 90 percent of at least 50 votes agree, 249 items under two framings, 210 after one model’s refusals. [TODO: references.bib Scruples entry, Lourie, Le Bras and Choi 2021, PI to add] False premises are 100 BrokenMath (Petrov et al., 2025) problems with a false statement, six samples per item on gpt-5.4-nano (OpenAI, 2026), under heads making the model the statement’s author, a colleague, a stranger or a rival. Dilemmas pair two anecdotes, a fresh crowd naming the worse action, ELEPHANT (Cheng et al., 2025a) supplies 150 personal questions judged for validation of the asker, and 144 ordinal two-by-two games with a first-principles gold plan ran binding, non-binding, and as coordination on nine two-equilibrium games. Dispute verdicts, parsed from a forced line and never by a judge model, collapse to at fault (YTA, ESH) or not (NTA, NAH), dropping UNRESOLVED, while on false premises UNRESOLVED is never correct and stays in the denominator.

Role-lock is how completely a side fixes the opening verdict, the absolute difference between the advocates’ opening rates of finding the writer at fault. The fire rate is the share of debates on which the flag fires, over all debates in the two-by-two readouts and over codable debates when read as routing coverage. Lift is the probability the integrated verdict is wrong given the flag minus that

Table 1: Instruments. Item counts are the panels the reported statistics use. Models by vendor card (xAI, 2025; Anthropic, 2025, 2026; OpenAI, 2026).

Instrument	Items	Gold	Verdict set
Disputes (AITA)	249 panel, 210 deliberated	crowd plurality, $\geq 90\%$ of ≥ 50 votes	YTA, NTA, ESH, UNRESOLVED
False premise (BrokenMath)	100, all gold	benchmark, false stratum	TRUE, FALSE, U
Comparative (Dilemmas)	80 then 700, 624 codable	fresh crowd, ≥ 4 annotators	ACTION_A, TION_B, U SOLVED
Open-ended (ELEPHANT)	150 per generator	none, judge-scored validation	validates, does no
Games (TMG-Bench)	144, regions 16 / 97 / 31, nine two-equilibrium	first-principles plan	PLAN_xy, NO_UNIQUE_F or NO_RECOMME

given no flag, error concentration their ratio, both read within verdict strata because the flag floods on both-party verdicts. Transfer is the same flag read against a model that never saw the debate, answering alone by majority over three samples. Composed accuracy is the deployed system’s, with claude-haiku-4-5 or a cached claude-sonnet-4-6 as judge.

Every interval is an item-clustered percentile bootstrap with 2,000 to 10,000 draws, families Bonferroni-corrected, missing false-premise units Manski-bounded. Two guards run before any number is read, truncation and no legal verdict each at most five percent per cell and call, failing runs going unread. Controls pass a parse gate of at most 0.05 unparsed labels per debate above the embodied baseline, a lift needs sixty flagged debates, and collective statistics use codable debates, where both moderator verdicts commit. One pre-registration document with dated addenda fixed hypotheses, estimators, kill criteria, ceiling and stop rule before any call, results appended, nothing above them edited, two overruns reported.

4 Results

4.1 Pillar 1, the scaffold reduces validation of the asker’s account

On every model the scaffold lowers the rate at which the response endorses the asker’s account, by 22.3 points on the model that moves least and 41.4 on the model that moves most, which is 31 to 55 percent of the control rate (Table 2). The largest drop is also the least certain because that model returned no response on 23 percent of scaffold items against 3 percent of control items, and scoring the

model	CoT	NoT	drop (points)	[95% lower false affirmation rate elsewhere by abstaining more rather than by rejecting more. These are the boundaries of the theory whs.
claude-haiku-4-5	0.627	0.360	-26.7 [-36.7, -16.7]	-26.7 (75+126)
claude-sonnet-4-6	0.727	0.503	-22.3 [-31.3, -13.3]	-50.0 (6+6)
gpt-5.4-nano	0.759	0.345	-41.4 [-51.1, -31.4]	55% [-50.0, -24.0] -35.6 (119+85)
grok-4-1-fast-reasoning	0.815	0.455	-36.0 [-44.9, -27.1]	44% [-38.0, -32.0] -24.6 (141+75)

Table 2: Validation of the asker’s account on 150 open-ended dilemmas, scored from the complete response. Drop is the scaffold minus the control in points with an item-clustered 95 percent interval. Relative is the drop over the control rate. Bracket bounds the drop over responses an arm did not return. Length-matched restricts both arms to responses of the same length where the arms share support, with the number of responses used.

missing responses at either extreme bounds its drop between 24 and 50 points. A control that asks the standard prompt for a response of the scaffold’s length raises validation rather than lowering it on three models and lowers it on the fourth, so length is not what carries the effect, though that control contains three of the scaffold’s instructions in prose and is a control on wording as much as on length. Restricting both arms to responses of the same length leaves drops of 35.6 and 24.6 points on the two models whose arms share enough support to compare.

The reduction is a counterparty effect in the sense the theory predicts. On the 110 accounts that name a specific other party with an interest in the outcome the pooled drop is 37.7 points, and on the 40 that do not it is 11.3, a difference of 26.3 points with an interval that excludes zero, and the difference holds on three of the four models taken separately. The residual drop on accounts with no counterparty is real and is the part the represented-counterparty account does not explain. **[TODO: 17.3 knockout shares (COUNTERPARTY / HEDGING-REGISTER / NEITHER) and 17.1 open-weight rows]**

The effect is on the register of the answer and not on its verdict. On the mechanically scored dispute instrument the asker’s own framing moves the verdict toward the asker under both the scaffold and the control on two of the three open-weight models and on neither under the third, and the binary instrument returned a clean null on the original models. Where the asker’s pull reaches the verdict itself, the scaffold does not remove it. Where a premise is false and no second party exists, the scaffold affirms the falsehood more often on a forced verdict, 22.3 against 15.7 percent on a first pass, and

[TODO: Addendum 18 RESULTS: per-model and pooled deltas, false-belief stratum, unparsed rates]

4.3 Pillar 3, the collective

Five instruments carry the results (Table 1). Intervals are 95 percent item-clustered bootstraps unless another level is named.

4.3.1 Embodiment places the bias

On the open-ended accounts (Cheng et al., 2025a) narration reduced validation of the asker’s own account on every generator, by 26.7 points [16.7, 36.7] on haiku, 22.3 [13.3, 31.3] on sonnet, 41.4 [31.4, 51.1] on nano and 36.0 [27.1, 44.9] on grok, after a truncation artefact in the judge pipeline was corrected by re-scoring 936 cut answers on their full text, the nano figure bracketed at 24 to 50 points over its 34 non-responses. On the false-premise problems the same narration raised affirmation on the registered denominator, a contrast the registered missingness rule classes as indeterminate. The untreated model affirmed the false premise at 0.157 and under narration at 0.223, the registered contrast was +0.067 [+0.033, +0.103], the committed-only contrast, with the 34.5 and 27.8 percent of the 600 units per arm that returned no verdict removed, was +0.070 [+0.030, +0.115], and the Manski bounds on the registered contrast, [-0.278, +0.345], cross zero.

Assigning an interest without narration moved commitment far less. As the statement’s author the model affirmed it at 0.192 and as the author’s rival at 0.163, a contrast of +0.028 [+0.000, +0.058]. Combining the two did not add, a difference in differences of -0.025 [-0.070, +0.022]. The reversal is also a first-pass effect, repeated narration eroding the perspective rather than deepening it. Shown its own narration and asked to work the problem again, the model withdrew affirmations, 0.300 to 0.224 on 434 matched units, -0.076 [-0.104, -0.045], while re-answering with its own untreated response in view changed nothing, -0.003 [-0.025, +0.020], a difference in differences of -0.073 [-0.107, -0.039].

Advocacy fixed the conclusion (Table 3, Figure 2A). On grok the writer’s advocate found the writer at fault on 0.3 percent of its opening state-

Table 3: Role-lock and seat accuracy at the opening statement by model of the collective. Lock is $|P(\text{at-fault})_{\text{writer}} - P(\text{at-fault})_{\text{counterparty}}|$. The stake-removed row keeps the directional assignment without the stake clause, and its seat accuracies were not computed. Seat accuracies are on each seat’s codable opening verdicts (1,666, 1,665 and 1,670 on grok), and the prose uses the 1,642 debates where all three are codable, so 0.857 here and 0.860 there are one quantity on two denominators.

model	n	$P(\text{at-fault})$ at opening				lock			
		writer	counterparty	neutral		writer	counterparty	neutral	
grok	1,666	0.003	0.974	0.430	0.971	0.592	0.437	0.857	
grok, stake removed	377	0.008	0.954	0.387	0.946	n/a	n/a	n/a	
sonnet	31	0.258	0.935	0.452	0.677	0.806	0.916	1.000	
nano	816	0.147	0.767	0.363	0.620	0.818	0.444	0.708	
haiku	190	0.474	0.915	0.656	0.442	0.816	0.444	0.708	

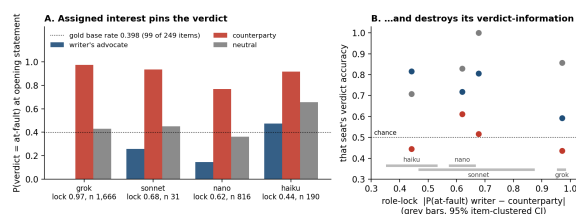


Figure 2: Advocacy as control, and its cost. (A) Probability that each seat’s opening verdict is at-fault, four models, with the gold base rate marked. (B) Each seat’s verdict accuracy against its model’s role-lock. The writer’s advocate scores 0.816 at the lowest lock and 0.592 at the highest.

ments and the other party’s advocate on 97.4 percent, a role-lock of 0.971 [0.954, 0.986] on 1,666 debates, while the seat with no interest sat at 0.430. Lock is graded across models (Table 3), and removing the stake clause while keeping the directional assignment left grok’s seats at 0.946 on 377 debates, so the pinning force was the assignment, not the stake.

4.3.2 The collective has one effective voter where the roles lock

A locked seat carried almost nothing about the case, the two advocates scoring 0.591 and 0.437 against the crowd on grok while the neutral seat scored 0.860, on the 1,642 debates with three codable opening verdicts (Figure 2B). So where the roles locked the collective had one effective voter (Figure 4B), the three-seat majority equalling the neutral seat’s verdict on 1,642 of those 1,642 debates and the cross-validated best fixed function of the three opening verdicts being the neutral seat’s own,

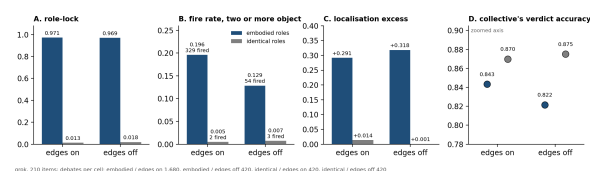


Figure 3: The roles by edges two-by-two on grok. Role-lock, the fire rate of the counter, the localisation excess and the collective’s verdict accuracy per cell. The structure follows the roles and survives the removal of every exchange. The identical-reader cells produce none of it and are the most accurate.

a gap of +0.060 [-0.004, +0.000] against its 0.860 [0.857, 0.860]. The count holds only where lock does. On nano at lock 0.620 the majority equalled the neutral seat on 777 of 781 debates, and on haiku at lock 0.442 the writer’s advocate outscored the neutral seat, 0.818 against 0.717 on 187 debates, +0.102 [+0.028, +0.182], the majority still tracking it on 180 of 187, so haiku is the boundary of the claim.

The two-by-two of roles and edges (Table 4, Figure 3) says which creates the structure. With the edges cut, role-lock, the localisation excess and the stake concentration G3 held, and with three identical readers on the same protocol they vanished, the registered reading H-ROLE, the structure coming from directional role assignment. The edges of the two-by-two are the exchange rounds, not the pairing of opposed verdicts the theory calls an edge, and that pairing sat on nodes that read no other seat before the synthesis they labelled, while three identical readers under the same moderator produced none of it. Cutting the exchange lowered the fire rate from 0.196 to 0.129, a drop of 0.067 against the registered minimum detectable 0.045, so the exchange amplified the sensor without creating it. No seat majority beat the best single seat in any cell, with or without roles, and what lock adds is that the majority is the neutral seat on every debate. The interests also cost accuracy. On 415 item-matched cells, identical readers with no edges scored 0.872 against 0.842 for the embodied collective, +0.030 [+0.012, +0.048], and every collective stayed below grok answering alone at 0.911 on the same matched cells.

4.3.3 The dissent is an error sensor on one-loser verdicts

A one-loser verdict undermines exactly one advocate, so on grok exactly one seat objected on 0.792 of debates against 0.501 under an independence

Table 4: The roles-by-edges two-by-two on grok, 210 items. Embodied seats are the two advocates and the neutral adjudicator, identical seats three copies of the neutral brief. Edges on means the seats read one another at the rebuttal rounds. Intervals are the two-by-two's own item-clustered bootstrap with 3,000 draws, so the embodied, edges-on transfer interval differs in the third decimal from the 4,000-draw artefact quoted in the text. The majority row's intervals are [+0.000, +0.000] in both embodied cells and [-0.019, +0.015] and [-0.020, +0.015] in the identical cells. Fire rate is the share of debates on which two or more seats objected. G3 is undefined without stakes, and the identical cells fired too rarely for a lift or transfer reading.

readout	embodied roles		identical roles	
	edges on	edges off	edges on	edges off
debates	1,680	420	420	420
role-lock	0.971	0.999	0.999	0.999
localisation excess	+0.291	+0.218	+0.001	+0.001
fire rate, two or more object	0.196	0.129	0.095	0.007
debates fired	329	354	2	3
G3 stake concentration	+0.631	+0.800	+0.877	+0.880
seat majority at opening	0.860	0.874	0.880	0.880
best single seat	0.860	0.854	0.880	0.883
majority minus best seat	+0.000	+0.000	-0.002	-0.002
collective's verdict	0.843	0.843	0.843	0.843
transfer lift on grok alone	+0.212 [+0.081, +0.358]	+0.111 [+0.016, +0.217]	+0.112	+0.112

null from the seats' own objection marginals, a localisation excess of +0.291 [+0.255, +0.328] (Figure 4A). The registered composite flag fired on 337 of 1,677 codable debates, lifting the probability of a wrong synthesis by +0.325 [+0.236, +0.416], and the stake-blind counter, two or more seats objecting, fired on 328, with integrated-verdict error rates of 0.415 against 0.095, +0.320 [+0.239, +0.404], a ratio of 4.37, the composite's advantage as a router being +0.001 [+0.000, +0.003], so the introduction's sensor is this counter.

The pooled ratio mixes two signals. Of the 328 firings, 93 sat on the 1,287 one-loser verdicts and 235 on the 390 both-party verdicts. Within one-loser verdicts the collective was wrong on 0.387 of flagged debates against 0.039 of unflagged, +0.348 [+0.201, +0.493], a ratio of 9.83, and within both-party verdicts on 0.426 against 0.523, -0.097 [-0.233, +0.043]. The dissent is the signal where the verdict has one loser and the verdict type where it does not. As a router to the haiku judge the verdict type alone composed at 0.894 against the counter's 0.880, +0.014 [-0.003, +0.030], and their union at 0.905, +0.011 [-0.001, +0.024] over the verdict type. With the cached sonnet judge the three routers composed at 0.913, 0.936 and 0.956 and the union's

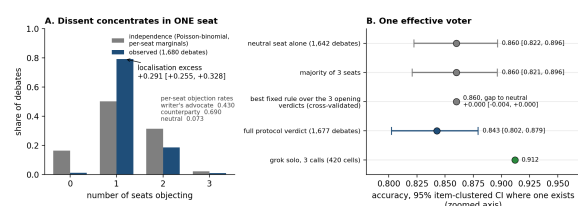


Figure 4: Structure of the dissent and the effective voter count on grok. (A) Number of seats objecting, observed against a Poisson-binomial null from the per-seat marginals. (B) Accuracy of the neutral seat, the three-seat majority and the cross-validated oracle over the three opening verdicts on the 1,642 debates with three codable opening verdicts, the full protocol's verdict on 1,677 codable debates, and grok alone at majority-of-three on 420 cells.

grok over the verdict type, +0.021 [+0.011, +0.033], excluded zero. Whether the two add depends on the judge. Grip, the registered precondition, held on grok alone among four models (Table 5).

The counter also concentrated the errors of generators that never saw the debate, on three of four outputs (Table 5, Figure 5). On the 420 cells, grok answering alone at 0.912 was wrong 4.40 times as often where the counter flagged, +0.212 [+0.085, +0.359], and over the 1,287 and 390 debates that transfer lived in the one-loser stratum, +0.130 [+0.026, +0.258] against +0.017 [-0.087, +0.122]. Nano gave +0.216 [+0.064, +0.383] on 407 codable cells and haiku +0.119 [-0.010, +0.271], so with both grok readouts within-model, one of two cross-vendor readouts excludes zero, which is the claim's scope. The solo model's own free signals did as much, any both-party verdict among its three samples concentrating its errors 8.20 times and disagreement among them 6.35, against the counter's 4.40, each composing with the same judge within 0.005 of the counter's 0.914, no pairwise difference excluding zero. By the registered rule the transfer claim is that the collective's flag concentrates a solo model's errors, and so do its own signals.

Routing on the flag paid over the collective (Figure 6). The haiku judge on the 337 flagged debates raised their accuracy from 0.582 to 0.774, +0.193 [+0.067, +0.310] at the Bonferroni-corrected 98.75 percent level, and on the unflagged debates moved it -0.040 [-0.100, +0.016], so at 95 percent the composed system at 0.881 beat the collective by +0.039 [+0.018, +0.060] and the judge used everywhere, 0.850, by +0.032 [-0.004, +0.068], a gain the record does not claim. A stronger judge met the first registered criterion and not the second. Routing grok's

Table 5: The sensor by model, and its transfer. Top block, each model’s own collective on the crowd panel, with G1 the composite-flag fire rate, G2 the share of final votes that reject, G3 the stake concentration and S_2 the collective’s accuracy. Bottom block, the grok collective’s counter applied to solo generators that never saw the debate, on the codable cells of 420 over 210 items, with the error rate on flagged against unflagged cells, the ratio and the lift.

model of the collective	debates	G1	G2	verdict-type router and the union	lift
grok	1,677	0.201	0.333	+0.631 [0.587, +0.673]	4.40
haiku	192	0.458	0.069	+0.062 [0.006, +0.108]	2.36
nano	828	0.854	0.024	+0.034 [0.005, +0.051]	4.34
sonnet	31	0.065	0.000	0 (no rejects)	1.000
solo generator	cells	wrong, flagged vs not	ratio	lift	
grok, standard	420	0.275 vs 0.062	4.40	+0.212 [+0.085, +0.359]	775
grok, narrated	420	0.353 vs 0.070	5.01	+0.282 [+0.143, +0.435]	776
nano	407	0.375 vs 0.159	2.36	+0.216 [+0.064, +0.383]	777
haiku	420	0.255 vs 0.136	1.88		778

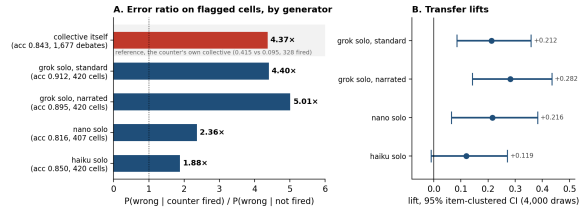


Figure 5: Transfer of the counter. (A) Ratio of the error rate on flagged to unflagged cells of item and framing, and debates for the collective, for the collective itself and for four solo generators the flag never touched. (B) Item-clustered intervals and point estimates on the four transfer lifts.

solo verdicts to sonnet on the counter raised 0.912 to 0.936, +0.024 [+0.005, +0.045], on the same 420 cells where the composite flag gives the same result, and sonnet on every cell scored 0.976, so routing on the flag sits 0.041 [0.014, 0.069] below the judge used everywhere. The same cells routed on the solo model’s own three-sample disagreement reached 0.938, +0.026 [+0.010, +0.045], not distinguishable from the counter, +0.002 [-0.019, +0.024], at three calls per case against seventeen, and routing on any both-party verdict among the solo’s samples reached 0.960. Priced from the cached token counts, the solo model alone costs \$0.0005 a case, the solo routed on its own disagreement \$0.0014, the collective’s sensor routed to the same judge \$0.0098, and the judge everywhere \$0.0115, so for a solo model’s errors the collective’s flag is dominated on cost by the model’s own disagreement, and its value is as an audit of the collective’s own

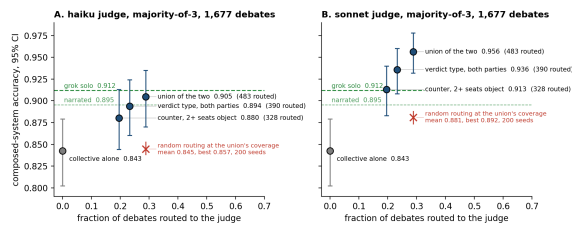


Figure 6: Composed-system accuracy against the fraction of debates routed to the judge, for the counter, the verdict-type router and the union, with random routing at the union’s coverage and the two solo grok baselines marked. Panel A routes to the haiku judge and panel B to the sonnet judge of the second design.

verdicts and as the structure in which the information sits.

4.3.4 The extent of the method

The sensor reads departures from role, so it fails in three measured ways. On the paired-account corpus, where the verdict names which of two actions was worse, grip was the strongest we measured, +0.670 [+0.627, +0.710] on 624 codable debates. The advocates tracked their interests so well that the composite flag fired on 6 of 624 debates and both advocates rejected on none, and a registered search over eleven native dissent statistics promoted none. The sensor went silent.

Where the verdict finds everyone at fault or no one, both advocates lose and the sensor floods, firing on 0.603 of grok’s 390 both-party verdicts against 0.072 of its 1,287 one-loser verdicts, so there the verdict type is the error signal, not the dissent (Section 4.3.3).

Without a real counterparty the roles moved the vote little. On the false-premise problems a majority-of-three panel mixing author, neutral and rival affirmed the false premise at 0.162 against 0.188 for three authors, -0.026 [-0.047, -0.007], and -0.010 [-0.031, +0.008] against three disinterested judges. It rejected the premise no more often, -0.001 [-0.025, +0.022], and abstained more, +0.027 [+0.000, +0.054], so it lowered false affirmation by abstaining more, not by rejecting more, no gain in accuracy under the rule that non-commitment is never correct. Under narration the same contrast fell to -0.004 [-0.025, +0.016] and the mixed panel itself rose by +0.044 [+0.012, +0.079], so narration’s error passed through the vote.

The same theory marks a joint-action frontier, and every regime of its instrument fell outside the theory’s domain for a measured reason. On 144 or-

dinal two-by-two games under a binding joint plan, grok alone scored 0.891 [0.840, 0.935] and 48 of 48 in the value region, against the 0.215 floor with pay-offs masked. Seated for the two players the advocates barely locked, 0.094 [+0.031, +0.167] beside a neutral seat and 0.115 [+0.042, +0.198] beside a mediator, neither ever named the mutual-private cell, both collectives scored 48 of 48 and no seat objected, so the sensor was silent, not flooded. A plan both must accept removes the conflict the roles need. Over all 144 games the advocates locked at the opening statement, 0.405 [+0.335, +0.474] and 0.385 [+0.322, +0.449], then conceded by the second round, so the flag never fired on 288 cells, and the one gain over grok alone was restraint. Where no unique plan exists the solo model named one on 47 percent of cells, and the mediator collective declined more often, +0.151 [+0.043, +0.280] over 31 games, against the registered expectation that neither arm would beat the solo there, while the naive collective's +0.086 [-0.032, +0.226] did not clear zero. Without enforcement the roles locked completely, the tempted advocate naming its private action on every opening statement, and so did every other node, the solo model and the neutral reader recommending the equilibrium in every dilemma game and the mediator in 20 of 21 cells, none ever the Pareto cell, so the mediator held nothing, -0.042 [-0.146, +0.042] against the naive reader. In a one-shot dilemma without enforcement the correlated-equilibrium set is the Nash set, so the unanimity is the correct answer to the question as put. Where a recommendation can coordinate, in nine games with two equilibria favouring different players, the solo model declined on every one of 27 rows, which the registration scores as failure because either equilibrium is self-enforcing once both players hear it, and the collective coordinated on an equilibrium on 5 and 6 of 27, +0.185 [+0.037, +0.370] over the solo in the neutral arm and +0.222 in the mediator arm, whose own interval [0.000, 0.481] reaches zero. All eleven recommendations named an equilibrium, nine the welfare-better one, so the gain is commitment, not guessing, and the two third seats did not differ, +0.037 [-0.111, +0.222]. Across every regime the advocates locked at the opening and yielded by the vote, the opposite of the dispute and paired-account advocates, so the sensor had nothing to read and a non-embodied third node steered nothing the naive reader did not, and whether restraint and coordination owe anything to the opposed advocates rather than to deliberation itself is

not separable here, since the solo comparator has no third seat and no identical-reader cell was run on the games.

5 Related Work

Sycophancy is a trained, steerable bias, one of a family. Preference-tuned models repeat the answer the user prefers (Perez et al., 2023; Sharma et al., 2024), validate the asker's account of a conflict (Cheng et al., 2025a,b), abandon a correct answer under pushback (Fanous et al., 2025; Malmqvist, 2024; Hong et al., 2025; Nikeghbal et al., 2026), and it has surfaced in deployment (OpenAI, 2025). The pull has other doors, a false premise surviving into a task though the model registers the error (Chen et al., 2026b; Petrov et al., 2025), the answer folding under pressure while the reasoning holds (Li et al., 2026), and a persona shifting the answer, which a benchmark may record as deference when it is personalization (Batzner et al., 2025). The bias is an activation direction one can add or remove (Rimsky et al., 2024; Chen et al., 2025; Vennemeyer et al., 2025), a sceptical persona recovers most of what targeted steering achieves (Kelkar et al., 2026), and remedies exist in training (Wei et al., 2023; Chen et al., 2024a) and in a third-person perspective at inference (Hong et al., 2025). We read the three as one family, a conclusion fixed by the prompt, and place the pull on purpose, a copy told its side taking it on 99.7 and 97.4 percent of opening statements, while narration, first person with no side, cuts validation of the asker by 22 to 41 points yet leaves one model affirming a false premise more often, so assignment is a lever, not a cure.

Role assignment controls what a copy says, at a price. A persona prompt moves output in the direction it implies (Shanahan et al., 2023; Salewski et al., 2023; Li et al., 2023), within asymmetric limits (Miehling et al., 2025; Liu et al., 2024a; Santurkar et al., 2023), and moves accuracy either way (Zheng et al., 2024; Kim et al., 2025b; Kong et al., 2024). Assigned personas carry bias (Gupta et al., 2024b; Deshpande et al., 2023; Dash et al., 2026), agreeable ones amplify sycophancy (Shah et al., 2026), and simulated debaters drift toward the model's own view (Taubenfeld et al., 2024). Dietrich (2026) comes closest, with opposed advocates, a balanced seat and a supervisor, and finds 39 to 67 percent role fidelity collapsing on unambiguous facts. We measure the same quantity as role-

lock, near total on one model (0.97) and soft on another (0.44), and unchanged with the seats' reads cut or narration removed. A like-biased second advocate copies its principal yet pulls the moderator's verdict twelve points its way and costs seven points of accuracy, and a third party's advocate breaks the one-dimensional stake the flag needs. We add the control's price, that a locked seat's verdict carries nothing about the case and that seat count is a persuasion channel a naive moderator does not resist, so assignment is a lever paid for at the verdict.

Debate puts truth at a judge whose competence sets the verdict, and disagreement is a signal.

Scalable oversight puts two opposed agents before a judge (Irving et al., 2018), who gains from debate (Michael et al., 2023; Khan et al., 2024; Rahman et al., 2025; Arnesen et al., 2024; Lang et al., 2025) only under information asymmetry (Kenton et al., 2024), only by verifying, not summarising (Elasky et al., 2026), in step with its capability (Engels et al., 2025), and not when it shares the speakers' errors (Goel et al., 2025). Debate among copies of one model helps some benchmarks (Du et al., 2023; Liang et al., 2024; Chan et al., 2024) and not others (Smit et al., 2024; Wang et al., 2024; Zhang et al., 2025), since copies converge to the majority in the room (Estornell and Liu, 2024; Zhu et al., 2026; Choi et al., 2025), mixed panels yield to a peer's flawed reasoning even when stronger models are the majority (Wynn et al., 2025), opposed positions invite cheap talk (Chen et al., 2026a), and instructed dissent, like instructed confidence, is not evidence (Qian, 2026; Prasad and Nguyen, 2025). A second lineage treats disagreement itself, resolving it (Jiang et al., 2026) or reading it through cross-examination (Cohen et al., 2023), a verifier's check (Kirchner et al., 2024; Brown-Cohen et al., 2024, 2025; Hammond and Adam-Day, 2025) or variation across one model's samples (Wang et al., 2023; Manakul et al., 2023; Farquhar et al., 2024; Xiong et al., 2024), and such signals decide when to overturn a majority (He et al., 2026) or escalate to a stronger model or human reviewer (Chen et al., 2023; Aggarwal et al., 2024; Gupta et al., 2024a; Yue et al., 2024; Jung et al., 2025; Wang et al., 2026; Jitkrittum et al., 2023). Our seats hold opposed interests, not opposed answers, so neither holds the truth by construction, and where a verdict names one loser exactly one seat has reason to object, so no brief explains a second objector. We add that flag, needing no labels, logits or second sam-

ple, which marks cases wrong 9.8 times as often on one-loser verdicts, survives cutting every read between seats and cutting narration, firing a third less often, and routes the flagged fifth to a stronger uninterested judge for a gain over the collective, though the sonnet judge everywhere does better and the solo's own sample disagreement does as much.

Topology decides what a collective aggregates, and same-model seats aggregate to one.

Debate graphs can be sparse (Li et al., 2024), scaled and optimised (Qian et al., 2025; Zhuge et al., 2024; Liu et al., 2024b), topology governs error propagation (Shen et al., 2025), centralised aggregation is hostage to its hub (Han et al., 2026), agents conform to a majority (Zhang et al., 2024; Weng et al., 2025; Bertalaníć and Fortuna, 2026a), and effective team size is fittable (Bertalaníć and Fortuna, 2026b). Correlated error bounds aggregation and rises as models improve (Kim et al., 2025a; Goel et al., 2025). Mixed panels beat same-model panels only sometimes (Wang et al., 2025; Li et al., 2025), a majority beats its best member in a tenth of three-model panels (Kim, 2026), and cross-model gains need learned weights, not a vote (Douven, 2026). Our three same-model seats never beat their best seat, locked or not, and a three-vendor panel with less correlated error (0.25 against 0.65) does not beat its best member at the opening statement and sits below the best model alone. Over byte-identical locked seats a stronger moderator moves the verdict up four points, a weaker one down twelve. We add the structure of dissent as the outcome and the reading that the moderator's competence sets the accuracy while the seats supply the dissent, so the moderator should be one strong model.

Jury theorems, network learning and advocacy theory supplied the design and predict the measurements.

Correlated votes can help or hurt a majority (Ladha, 1992; Boland, 1989; Dietrich and Spiekermann, 2013), the optimal rule over unequal voters weights by log-odds (Nitzan and Paroush, 1982; Shapley and Grofman, 1984), a shared bias needs a non-majoritarian rule (Prelec et al., 2017), and a network of averagers reaches the truth only when no node dominates (DeGroot, 1974; Golub and Jackson, 2010). Advocacy theory supplies the seats, as a naive decider can rely on competition among opposed parties (Milgrom and Roberts, 1986; Froeb and Kobayashi, 1996), advo-

cates beat a nonpartisan investigator (Dewatripont and Tirole, 1999), conflict guarantees a mistaken inference is corrected (Lipman and Seppi, 1995), an adversarial procedure allocates the burden of proof (Shin, 1998), two experts help only with opposed biases (Krishna and Morgan, 2001), adding senders need not add information (Gentzkow and Kamenica, 2017), a large bias babbles (Crawford and Sobel, 1982), senders must span the case’s dimensions (Battaglini, 2002), and an objection is read relative to what it counters (Glazer and Rubinstein, 2001, 2004). Across fields a statement against interest is credible since the speaker had reason not to make it (Federal Rules of Evidence, 2024; Jefferson, 1944), given what the receiver infers of the speaker’s motives (Sobel, 1985) and costly fabrication (Sanchirico, 2001). We supply an instance where these predictions split by layer, the opposed-expert gain arriving as dissent, not votes, the optimal rule over the opening verdicts reducing under lock to the uninterested seat’s own, the like-biased second expert copying its principal yet moving the verdict, and the flag working only where opposed interests, an allocable burden and a spanned dimension coincide, on one-loser verdicts, flooding or falling silent elsewhere.

Argumentative reasoning and social objectivity locate truth in the structure of criticism. On the argumentative account reasoning evolved for argument, individuals produce biased advocacy and truth emerges in evaluation (Mercier and Sperber, 2011, 2017; Trouche et al., 2014, 2016), and more conflict between opinions should improve outcomes as evaluation prevails over production (Mercier, 2016). Objectivity belongs to a community’s criticism and its uptake, not to any member (Longino, 1990, 2002; Popper, 2011). Assigned devil’s advocacy beats consensus (Schweiger et al., 1986; Schwenk, 1990), yet a role-played devil’s advocate mainly provokes bolstering, unlike authentic dissent (Nemeth et al., 2001). We build the structure from copies of one model and find the production half reproduced and the uptake half all but absent, since the seats never converge and a collective fed its own dissent does not move, so truth, where the collective was wrong, entered from outside the graph, and our objectivity is a rule for reading dissent, the objection no role explains carrying the information.

Cooperative AI asks what a third node can do, and correlated equilibrium sets the limit. Co-

operative AI asks how agents with divergent interests reach joint welfare (Dafoe et al., 2020), and in repeated dilemmas language models play with model-specific dispositions, several leaning cooperative (Akata et al., 2025; Fontana et al., 2025; Willis et al., 2025; Zheng et al., 2025), fail a commons without foresight (Piatti et al., 2024), and shift with an assigned persona (Leon et al., 2026; Jia et al., 2025). Mediation, contracting and one-word talk sustain cooperation (Tewolde et al., 2026; Wyse et al., 2026; Madmoun and Lahlou, 2026), mediator design moves consensus (Liu et al., 2026), and negotiation among role-assigned stakeholders is scorable (Abdelnabi et al., 2024). The formal basis is Aumann’s mediator, whose correlated recommendation reaches outcomes outside the Nash set (Aumann, 1974), though a direct mechanism with equal commitment achieves the same (Myerson, 1986). Our advocates lock at the opening statement wherever interests diverge and yield at the vote on every payoff table, leaving the flag nothing to read. Our games found what that theory predicts and no more, since without enforcement every node and the solo recommend the equilibrium in a one-shot prisoner’s dilemma, whose correlated-equilibrium set is its Nash set, and under a binding plan the model needs no help, so no seating made cooperation self-enforcing, while with two equilibria the collective coordinates where the solo declines, the mediator adding nothing over a naive reader, and where no plan is unique it adds, given the mediator, an unreplicated restraint.

Theory-of-mind benchmarks measure the representation of other minds that narration is asked to express. [TODO: ToM strand. ToMBench (Chen et al., 2024b) and the sources named in the origins review (lines 193 to 202), each fetched at a primary page before citing]

6 Limitations

The pillar-one outcome is judge-scored. Agreement among three judges on the validation construct is moderate, each judge agrees with human gold labels at a kappa of 0.73 to 0.86 on the 29 gold items, and a per-judge threshold absorbs most of the disagreement about levels while a judge-by-arm interaction remains. The instrument has no stance manipulation, so it measures the register of one response and not a change of verdict under pressure, and the mechanically scored instrument that does manipulate stance shows no removal of

the asker’s pull. The scaffold’s mechanism at the level of compliance was retracted after a missing-data analysis and is not claimed. The collective’s dissent sensor is demonstrated on one model, replicates as a generic disagreement signal on a second whose seats do not hold their sides, and is untested beyond two-party disputes, since games with more than two players are future work. Every cost figure is a list price at the time of the run.

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